PAYTON JOHNCOUR

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EDUCATION

University of California, Irvine — B.S. Computer Engineering (Transfer) Expected May 2029

Orange Coast College — A.S. Computer Science, Physics, Mathematics · GPA 3.8 May 2027

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Engineering Circuits

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Bash

Frameworks & Libraries: Next.js, React, Flask, PyTorch, OpenCV

Tools & Platforms: Git/GitHub, Visual Studio Code, Unix/Linux, Vercel, KiCad

Concepts: Data Structures & Algorithms, Object-Oriented Design, Computer Vision & Machine Learning, Embedded Systems, Full-Stack Web Development

PROJECTS

Bintelligence — Autonomous Waste Sorting System · Team Lead

bintelligence-front-end.vercel.app

- Designed, built, and tested a fully autonomous waste-sorting device achieving 80% classification accuracy across trash, compost, and recycling, removing the need for manual sorting downstream in the recycling process
- Trained a YOLOv11 object detection model in PyTorch on environment-specific datasets to identify waste categories in real time
- Implemented an event-driven control loop in C++ that converted sensor input into motor output, automating the full sorting pipeline
- Optimized computer vision models for edge deployment on a Sony IMX500 sensor, reducing inference latency and offloading compute from the host CPU

Math Solver — AI-Powered Problem Solving Web App

github.com/johncourpayton/MathSolver

- Built and deployed a full-stack web application in Next.js, Python, and Flask that automates recognition and step-by-step solving of mathematical formulas
- Integrated OCR and LLM APIs (Pix2Text for image-to-LaTeX extraction, Google Gemini for solution generation) into a single end-to-end pipeline
- Live-demoed the app to a judging panel, solving multi-step math and physics problems (capture → OCR → LaTeX → solution) in under 60 seconds despite chaining OCR extraction with LLM reasoning
- Won 1st place out of 25 submissions and a $500 scholarship for Best Overall Project

OC Robotics Rover — Embedded Software Developer

- Implemented an iterative acceleration-ramping algorithm in C++ that incrementally adjusted motor velocity setpoints, replacing abrupt on/off control and reducing mechanical stress on the drivetrain
- Collaborated with a team via GitHub, using issue tracking to identify, assign, and resolve software bugs

LEADERSHIP & ACTIVITIES

Micro Mouse @ OCC — Founder Jan 2026 – Present

occmicromouse.space

- Founded a student organization preparing teams for autonomous robotics competitions, growing active membership to 20+ students
- Lead weekly technical workshops on microcontroller programming, sensor integration, and maze-solving algorithms (wall-following through Flood Fill), building members' core CS and software engineering skills

WORK EXPERIENCE

Golf Services — Mesa Verde Country Club, Costa Mesa, CA June 2023 – Present

- Provide customer service and course-operations support in a fast-paced environment while balancing a full course load
- Coordinate with team members on daily logistics and equipment upkeep, reinforcing teamwork and time-management skills